ChE 381: Unit Operations Laboratory

Viscosity of Liquids

rev: Fall 2018

Introduction

Viscosity is a basic fluid property which is often required by engineers to make estimates of transport behavior such as mass transfer and heat transfer. It appears in many of the engineering correlations (e.g. Reynolds number). There are numerous compilations of viscosity values (see Reid, Sherwood, and Prausnitz [1]) for pure fluids and techniques to estimate viscosities for mixtures.

Theory [1], [2]

Viscosity is a fluid property and is defined as the ratio of the shearing stress on a fluid to the velocity gradient established in the fluid at the point of shear. The viscosity is a measure of the resistance of a fluid to a shearing force, or equivalently, the viscosity may be interpreted as a measure of the momentum flux in a fluid. This will be made clearer in the discussion presented below.

The general relationship[1,2] between the stresses in a fluid element and the velocity gradients is quite involved because the stresses in a fluid can only be correctly described using a second-order tensor. However, for one dimensional flow and for incompressible fluids, the relationship between the shear stress and velocity gradient in the fluid can be stated in a straightforward manner. (Note that gases in many cases can be treated as an incompressible fluid provided there is no large pressure gradient in the flow field. Gas flow, in a pipe or past a solid body at conditions typically encountered in the process industries can usually be treated as incompressible.) Consider a fluid constrained between two horizontal plates, as shown in *Figure 1*. (In practice, this is often achieved by using concentric cylinders, which have diameters that are large relative to the gap between the cylinders.) If one plate is moved at a constant velocity, a velocity profile is established in the fluid, and it has been found experimentally that the force required to keep the plate in motion is proportional to the velocity gradient in the fluid. This relationship may be written as follows:

$$\tau_{yx} = -\mu \frac{dv_x}{dy} \tag{1}$$

Where τ_{yx} is the shear stress (i.e., force per unit area of fluid contacting the plate) in the +(positive)x direction on a surface with unit normal in the +y direction; dv_x/dy is the gradient in the y direction of the fluid velocity in the x direction; and μ is the fluid viscosity, which by definition is the ratio of the shear stress to the velocity gradient in the fluid normal to the shearing surface. The negative sign arises from convention; if the shear stress is taken to be positive (in the positive x direction), the velocity gradient established in the fluid is negative relative to the coordinate system used in *Figure 1*. As expected intuitively, the shear force causes the fluid to move in the direction of the force; however, the fluid is not all transported at an equal rate. Instead a velocity gradient is established in the direction normal to the surface at which the force is applied. As a result, the momentum of the fluid decreases the father a fluid element is from the surface. Thus, the viscosity may be interpreted as the ability of momentum to diffuse through the fluid to layers further removed from the surface. Although the fluid momentum decreases with distance from the plate, the shear stress acting on each fluid later is the same. By Newton's second law of motion, a force equal but opposite in direction to that applied to the bottom plate is necessary to hold the top plate in position. For a given shear stress, a large velocity gradient will be established for a low viscosity fluid (i.e., the plate will move at a high velocity), whereas for a viscous fluid, the velocity gradient will be smaller since the plate will not move fast.

It was previously mentioned that the viscosity is a fluid property, which depends on the particular material and which must be individually measured for each substance. Experimentally, however, it has been found that fluids can be categorized into a number of general classes. Newtonian fluids are those that obey *Equation (1)*. For these fluids, and increase in the shear rate causes a proportional increase in the velocity gradient (i.e., μ is a constant). Many single phase, non-polymeric fluids are Newtonian. These include: all gases, water, oils and other hydrocarbon fluids, most molten metals, and many true solutions and mixtures. Non-Newtonian fluids, on the other hand, are fluids that do not exhibit a linear relationship between the shear stress and the velocity gradient. In these fluids, the apparent fluid viscosity will be a function of the rate of shear. The relationship between the shear stress and the velocity gradient shows a wide range of behavior for non-Newtonian fluids. Some examples are presented in *Figure 2*. Non-Newtonian fluids include: latex paints, polymer solutions, colloidal mixtures, foams, and two-phase fluids. Curves B and C in *Figure 2* are for fluids that are called pseudo-plastic and dilatant, respectively. For these fluids, a power-law relationship (also known as the Ostwald-deWaele Model) can adequately characterize the dependence of the shear rate on the velocity gradient:

$$\tau_{yx} = \pm m \left| \frac{dv_x}{dy} \right|^n \tag{2}$$

Here m and n are empirical constants and the $\pm$ sign is chosen such that the shear stress is in the correct direction relative to the velocity gradient. This equation is often written in the form:

$$\tau_{yx} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy} \tag{3}$$

Where μ_{app} is the apparent viscosity. Note that the apparent viscosity is not a constant but instead depends on the rate of shear experienced by the fluid. (The actual relationship between the velocity gradient in the fluid and the shear stress is given in terms of the two constants, m and n defined by *Equation (2)*.) For $n > 1$, the fluid behavior correcponds to curve C of *Figure 2*, whereas for $n > 1$, the fluid behavior is shown by curve B. For $n = 1$, *Equation (1)* is obtained with $m = \mu$.

Curve D shows the behavior of a Bingham fluid. This type of fluid remains stationary until a critical minimum shear stress is applied; once flow is initiated the fluid exhibits Newtonian behavior. Thick pastes are often Bingham fluids. Curve A is for a Newtonian fluid and shows the linear relationship between velocity gradient and shear stress.

To determine the viscosity of a fluid, it is necessary to measure the shear force applied to the fluid and the velocity field set up in the fluid. Fundamental studies are difficult to make. Experimental measurements of viscosity, though, are readily made. In such measurements, the equations of motion are solved, usually for a simple flow geometry, to determine the relationship between the fluid viscosity and readily measured macroscopic variables such as the flowrate and the pressure drop. The fluid viscosity is then obtained from the observed relationship between these measured macroscopic variables. For example, laminar flow is cylindrical capillaries is described by the Hagen-Poiseuille equation. By using this equation, one may determine the fluid viscosity from measurements of flowrate and pressure drop. In another method, Stoke's Law is used to relate the fluid viscosity to the drag force on a sphere moving through a fluid. The drag force on a sphere, in turn, can readily be obtained experimentally by measuring the free fall velocity of the sphere in the fluid. In this experiment, both of these methods will be used to determine the viscosity of a Newtonian and a non-Newtonian fluid.

Experiment:

The Capillary Tube Viscometer

In this experiment, the flowrate and pressure drop across a capillary tube is used to measure the viscosity of a fluid. The viscosity of the fluid is related to the flowrate and pressure drop by the Hagen-Poiseuille equation for a Newtonian fluid:

$$Q = \frac{\Delta P \pi R^4}{8\mu L} \tag{11}$$

Where Q is the volumetric flowrate of the fluid, ΔP is the pressure drop across the capillary, R and L the capillary radius and length, respectively, and μ the fluid viscosity. Thus measurements of Q vs. ΔP should yield a straight line of slope $\pi R^4/8\mu L$, from which the viscosity may be calculated.

An analytical equation for the flowrate in a tube can also be obtained for certain non-Newtonian fluids. For fluids that obey the Ostwald-deWaele viscosity model (*Equation (3)*), the following equation is obtained:

$$m\left[\frac{3n+1}{n}\right]^n \cdot \left(\frac{Q}{\pi R^3}\right)^n = \left(\frac{\Delta P}{2\mathrm{L}} R\right) \tag{12}$$

Where m and n are the parameters of the Ostwald-deWaele model and the other terms are the same as in *Equation (11)*. The derivation of the equation is given in appendix I.

A plot of $\log(\Delta P \cdot R/2L)$ vs. $\log(Q/\pi\, R^3)$ will give a slope of 'n' and an intercept value of $\log\left\{m\left[\frac{3n+1}{n}\right]^n\right\}$. From the slope and intercept values obtained from the graph, the parameters m and n can then be calculated as a function of capillary diameter and length.

Experimental Procedure

The apparatus consists of a small capillary tube attached to a small pressurization vessel. The capillary tube is removable and the tubes of different length to diameter ratios may be attached to the vessel. The vessel is filled with the fluid for which the viscosity is to be measured and pressurized with air. Use caution when pressurizing and depressurizing the vessel. Always open the vent valve carefully. Open the vent valve first when depressurizing the vessel; keep the vent valve open while filling the reservoir. Always close the vent valve last when pressurizing the vessel. A constant pressure may be applied to the vessel by adjusting the pressure regulator on the air line.

Experiments should be conducted with a non-Newtonian fluid. A non-Newtonian fluid may be prepared by mixing a **0.8 wt% solution** of CARBOPOL in warm tap water. Use the power mixer in the laboratory to prepare the solution. See supplemental information in Blackboard for more details on preparing Carbopol solutions.

Make experimental runs using several capillary tubes of different diameter and length. For each run, measure the flowrate at 4 or 5 different pressure differentials. The maximum pressure that should be used is 60 psig.

ANALYSIS:

Make plots of $\log(\Delta P \cdot R/2L)$ vs. $\log(Q/\pi R^3)$ and determine values for 'm' and 'n' for the different capillary tubes that you used. What are the units of dimension for 'm' and'n'? Do you obtain different results for capillaries having different L/D ratios? Why?

Some possible references for more information:

Aris, R., *Vectors, Tensors and the Basic Equations of Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, NJ (1962).

Whitaker, S., *Introduction to Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, NJ (1968).

Bird, R.B., Stewart, W.E., and Lightfoot, E.N., *Transport Phenomena*, John Wiley, New York (1960).

Batchelor, G.K., *An introduction to Fluid Dynamics*, Cambridge University Press, Cambridge (1967).

Perry, R.H. and Chilton, C.H., *Chemical Engineers' Handbook*, 5th ed., McGraw-Hill, New York (1973).

Dean. J.A., *Lange's Handbook of Chemistry*, 13th edition, McGraw-Hill, New York (1985).

References

[1] McCabe W.L., Smith J.C., Harriot P., *Unit Operations in Chemical Engineering,* 7th Edition pp. 46-52, (2005)

[2] R.C. Reid, J.M. Prousnitz, T.K. Sherwood, T*he Properties of Gases and Liquids*, 3rd ed. McGraw-Hill, 1977.

[3] R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed. John Wiley and Sons, Inc., 2002

[4] J. Davidson, R. Clift, and D. Harrison, Eds., *Fluidization*, 2nd ed. Academic Press, 1995.

[5] R.H. Perry and D.W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.

Figure 1. Velocity profile of a fluid contained between two parallel plates, one of which moving at a constant velocity v_0.

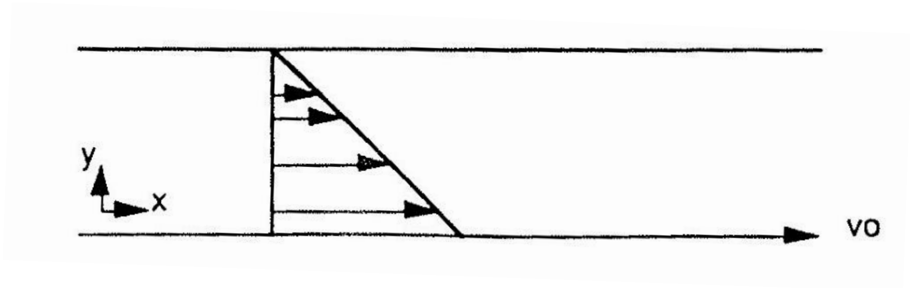
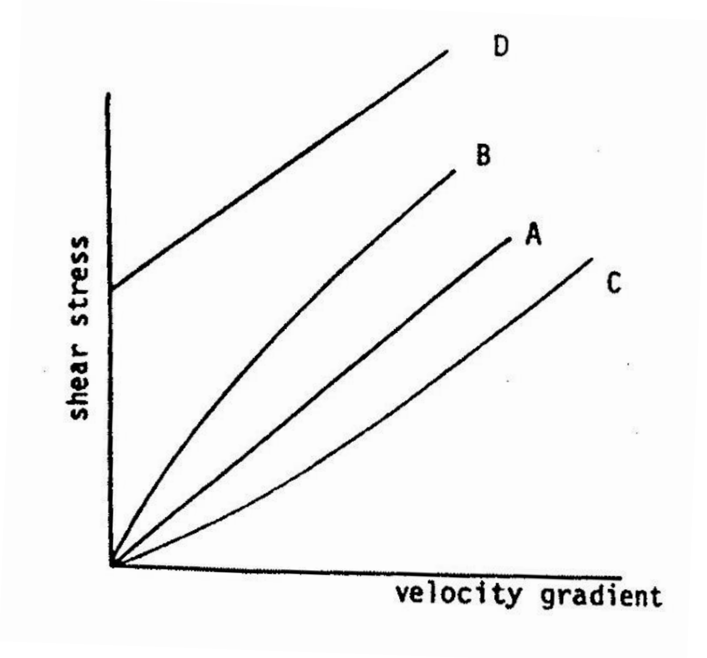


Figure 2. **Relationship between strain and shear for different fluids: (A) Newtonian fluid; (B) psuedo-plastic fluid; (C) dilatant fluid; (D) Bingham fluid.**



List of Symbols (SI units)

a = radius of the sphere (m)
F_b = buoyancy force (N)
F_D = drag dorce (N)
g = gravitational acceleration (m/s^2)
K = calibration constant
L = length of capillary (m)
m = mass (kg)
Q = volumetric flowrate (m^3/s)
R = radius of the capillary tube (m)
Re = Reynold's number (dimensionless)
t = time (s)
v = fluid velocity (m/s)
v_x = component of the fluid velocity in the x direction
ΔP = pressure drop (Pa)
μ = fluid viscosity (kg/m-s)
ρ_f = density of the fluid (kg/m^3)
ρ_s = density of the sphere (kg/m^3)
τ = shear stress (N/m^2)

Appendix I.

From the equation of motion in cylindrical coordinates and the equation of continuity for an uncompressible fluid one obtains:

(1)

$$\frac{dp}{dz} = -\frac{1}{r}\frac{d}{dr}(r\tau_{rz})$$

for flow in a cylindrical tube. Here z is the direction of flow, r is theradial coordinate in the cylindrical tube, dp/dz is the pressure drop across the tube and τ_{rz} is the shear stress in the z direction on a fluid surface with unit normal in the r direction. dp/dz, the pressure drop across the tube, is a constant at steady flow. This term is equal to $-\Delta P/L$ where ΔP is the pressure drop for a tube of length L.

Integration of (1) with respect to z gives:

(2)

$$\tau_{rz} = \frac{\Delta p}{2L} r$$

Next, one substitutes the power law model for τ_{rz} to obtain:

$$\begin{aligned}\tau_{rz} &= -m\left|\frac{dv_z}{dr}\right|^{n-1}\frac{dv_z}{dr}\\ &= m\left|\frac{dv_z}{dr}\right|^{n}\end{aligned}$$

since dv_z/dr has a negative slope in the coordinate system for this problem. Therefore one obtains:

$$\frac{dv_z}{dr} = \left(\frac{1}{m}\frac{\Delta P}{2L}\right)^{1/n} r^{1/n} = ar^{1/n}$$

Separation of variables, integration, and application of the no-slip boundary condition that: at $r = R$, $v_z = 0$ gives:

$$\int_0^{v_z} dv_z = a\int_R^f r^{1/n} dr$$

$$v_z = a\frac{n}{n+1} r^{\frac{n+1}{n}}\bigg|_R^r$$

$$v_z = a\frac{n}{n+1}\left(r^{\frac{n+1}{n}} - R^{\frac{n+1}{n}}\right)$$

The total flowrate, Q, is obtained by integration of the velocity profile over the tube cross section. In cylindrical coordinates this gives:

$$Q = \int_0^{2R}\int_0^R v_z\, r dr\, d\theta$$

$$= 2\pi a\left(\frac{n}{n+1}\right)\int_0^R \left(r^{\frac{n+1}{n}} - R^{\frac{n+1}{n}}\right) r dr$$

$$= 2\pi a\left(\frac{n}{n+1}\right)\left[\frac{n}{3n+1} R^{\frac{n+1}{n}} - \frac{1}{2} R^{\frac{n+1}{n}} R^2\right]$$

$$= 2\pi a R^{\frac{3n+1}{n}}\left(\frac{n}{n+1}\right)\left[\frac{n}{3n+1} - \frac{1}{2}\right]$$

$$= \pi a R^{\frac{3n+1}{n}}\left(\frac{n}{3n+1}\right)$$

Or $Q = \pi\left[\frac{1}{m}\frac{\Delta P}{2L}\right]^{1/n} R^{\frac{3n+1}{n}}\left(\frac{n}{3n+1}\right)$